# Beam Reinforcement Check Report

12. Structural Drawing_ SNIGDHOTARA.dxf | f'c=28.0 MPa | fy=420.0 MPa | 54 beams

### Cross-section checks: 73 total | PASS: 18 | FAIL: 55

## LONGITUDINAL SECTION OF FLOOR BEAM-FB1(12''x24''): ALONG GRID-7 3/7'' = 1'-0''

FB1 | 305x610mm (bxh) | 3 span(s) | 8 supports

#### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 7a-7a | 610 | 554.5 | d10 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |
| 7b-7b | 610 | 554.5 | d10 | 102 | 128 | PASS | 2/3 | PASS | FAIL | FAIL |

#### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=277mm (7a-7a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2962 | 2218 | PASS | 50 | 50 | PASS | 2962 | - | 1220 | PASS |
| 2 | 3564 | 2218 | PASS | 50 | 50 | PASS | 1200 | 1193 | 1220 | FAIL |
| 3 | 3668 | 2218 | PASS | 50 | 100 | PASS | 1200 | 1200 | 1220 | FAIL |

#### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 3 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 3 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 3 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 3 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-FB2(12''x24''): ALONG GRID-A 3/7'' = 1'-0''

FB2 | 305x610mm (bxh) | 2 span(s) | 6 supports

#### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 8a-8a | 610 | 552.5 | d12 | 102 | 128 | PASS | 2/3 | PASS | FAIL | FAIL |
| 8b-8b | 610 | 552.5 | d12 | 152 | 128 | FAIL | 2/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=152mm, d/2=276mm (8a-8a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4587 | 2210 | PASS | 50 | 50 | PASS | 1209 | 1209 | 1220 | FAIL |
| 2 | 3706 | 2210 | PASS | 50 | 50 | PASS | 1195 | 1195 | 1220 | FAIL |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 284 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 284 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 768 | 284 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 768 | 284 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

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## LONGITUDINAL SECTION OF FLOOR BEAM-FB2(12''x24''): ALONG GRID-B 3/7'' = 1'-0''

FB2 | 305x610mm (bxh) | 2 span(s) | 6 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 9a-9a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |
| 9b-9b | 610 | 552.5 | d12 | 152 | 128 | FAIL | 3/3 | PASS | PASS | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=152mm, d/2=276mm (9a-9a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4587 | 2210 | PASS | 50 | 50 | PASS | 1209 | 1209 | 1220 | FAIL |
| 2 | 3156 | 2210 | PASS | 50 | 50 | PASS | 3156 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 768 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

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## LONGITUDINAL SECTION OF FLOOR BEAM-FB2(12''x24''): ALONG GRID-C 3/7'' = 1'-0''

FB2 | 305x610mm (bxh) | 2 span(s) | 6 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 9a-9a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |
| 9b-9b | 610 | 552.5 | d12 | 152 | 128 | FAIL | 3/3 | PASS | PASS | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=152mm, d/2=276mm (9a-9a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4587 | 2210 | PASS | 50 | 50 | PASS | 1209 | 1209 | 1220 | FAIL |
| 2 | 3156 | 2210 | PASS | 50 | 50 | PASS | 3156 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 768 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-FB2(12''x24''): ALONG GRID-D 3/7'' = 1'-0''

FB2 | 305x610mm (bxh) | 2 span(s) | 7 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 11a-11a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |
| 11b-11b | 610 | 552.5 | d12 | 152 | 128 | FAIL | 3/3 | PASS | PASS | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=152mm, d/2=276mm (11b-11b) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4612 | 2210 | PASS | 50 | 50 | PASS | 1209 | 1209 | 1220 | FAIL |
| 2 | 2250 | 2210 | PASS | 50 | 50 | PASS | 2250 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 426 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 426 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 852 | 213 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 213 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 768 | 426 | PASS |

| | | | | |
|---|---|---|---|---|
| 2 | Mn,r+ >= Mn,r-/2 | 768 | 426 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 852 | 213 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 768 | 213 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-FB2(12"x24"): IN BETWEEN GRID 4~6 3/7" = 1'-0"

FB2 | 305x610mm (bxh) | 1 span(s) | 3 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 12a-12a | 610 | 552.5 | d10 | 102 | 128 | PASS | 2/2 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (12a-12a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1825 | 2210 | FAIL | 50 | 50 | PASS | 1825 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 568 | 284 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 568 | 284 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 142 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 568 | 142 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-FB2(12"x24"): IN BETWEEN GRID 3~4 3/7" = 1'-0"

FB2 | 305x610mm (bxh) | 1 span(s) | 3 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 13a-13a | 610 | 550.5 | d12 | 102 | 128 | PASS | 2/2 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=275mm (13a-13a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2250 | 2202 | PASS | 50 | 50 | PASS | 2250 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 568 | 284 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 568 | 284 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 142 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 568 | 142 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-FB1(12"x24"): ALONG GRID-E 3/7" = 1'-0"

FB1 | 305x610mm (bxh) | 4 span(s) | 10 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 14a-14a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |
| 14b-14b | 610 | 552.5 | d12 | 152 | 128 | FAIL | 2/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=152mm, d/2=276mm (14b-14b) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4612 | 2210 | PASS | 50 | 50 | PASS | 1209 | 1209 | 1220 | FAIL |
| 2 | 2258 | 2210 | PASS | 50 | 50 | PASS | 2258 | - | 1220 | PASS |
| 3 | 1829 | 2210 | FAIL | 50 | 50 | PASS | 1829 | - | 1220 | PASS |
| 4 | 2502 | 2210 | PASS | 50 | 50 | PASS | 2502 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 3 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 3 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 3 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 3 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 4 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 4 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 4 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 4 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

## LONGITUDINAL SECTION OF GRADE BEAM-GB1(14"x26"): ALONG GRID-1 3/7" = 1'-0"

GB1 | 356x660mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 660 | 579.5 | d10 | 102 | 128 | PASS | 3/2 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1320mm) | Midspan s<=d/2: s=102mm, d/2=290mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1977 | 2318 | FAIL | 50 | 50 | PASS | 1977 | - | 1320 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 400 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 400 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 400 | 150 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 400 | 150 | PASS |

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## LONGITUDINAL SECTION OF GRADE BEAM-GB1(14''x26''): ALONG GRID-2 3/7'' = 1'-0''

GB1 | 356x660mm (bxh) | 2 span(s) | 6 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 2a-2a | 660 | 577.5 | d10 | 102 | 128 | PASS | 3/2 | PASS | FAIL | FAIL |
| 2b-2b | 660 | 577.5 | d10 | 152 | 128 | FAIL | 2/2 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1320mm) | Midspan s<=d/2: s=152mm, d/2=289mm (2a-2a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1772 | 2310 | FAIL | 50 | 50 | PASS | 1772 | - | 1320 | PASS |
| 2 | 4531 | 2310 | PASS | 50 | 50 | PASS | 1297 | 1297 | 1320 | FAIL |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 568 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 568 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 568 | 192 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 568 | 384 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 568 | 384 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 568 | 192 | PASS |

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## LONGITUDINAL SECTION OF GRADE BEAM-GB1(14''x26''): ALONG GRID-3 3/7'' = 1'-0''

GB1 | 356x660mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 3a-3a | 660 | 575.5 | d12 | 102 | 144 | PASS | 2/2 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1320mm) | Midspan s<=d/2: s=102mm, d/2=288mm (3a-3a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1335 | 2302 | FAIL | 50 | 50 | PASS | 1335 | - | 1320 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 568 | 284 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 568 | 284 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 142 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 568 | 142 | PASS |

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## LONGITUDINAL SECTION OF GRADE BEAM-GB1(14"x26"): ALONG GRID-4 3/7" = 1'-0"

GB1 | 356x660mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 3a-3a | 660 | 575.5 | d12 | 102 | 144 | PASS | 2/2 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1320mm) | Midspan s<=d/2: s=102mm, d/2=288mm (3a-3a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2343 | 2302 | PASS | - | - | - | - | - | 1320 | N/A |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 568 | 284 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 568 | 284 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 142 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 568 | 142 | PASS |

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## LONGITUDINAL SECTION OF GRADE BEAM-GB1(14"x26"): ALONG GRID-5 3/7" = 1'-0"

GB1 | 356x660mm (bxh) | 2 span(s) | 7 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 4a-4a | 660 | 577.5 | d10 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |
| 4b-4b | 660 | 577.5 | d10 | 152 | 128 | FAIL | 2/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1320mm) | Midspan s<=d/2: s=152mm, d/2=289mm (4b-4b) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2212 | 2310 | FAIL | 50 | 50 | PASS | 2212 | - | 1320 | PASS |
| 2 | 4206 | 2310 | PASS | 50 | 50 | PASS | 1297 | 1297 | 1320 | FAIL |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 852 | 384 | PASS |

| | | | | |
|---|---|---|---|---|
| 1 | Mn,r+ >= Mn,r-/2 | 852 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 213 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 852 | 213 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 852 | 384 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 852 | 384 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 568 | 213 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 852 | 213 | PASS |

## LONGITUDINAL SECTION OF GRADE BEAM-GB1(14"x26"): ALONG GRID-6 3/7" = 1'-0"

GB1 | 356x660mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 5a-5a | 660 | 575.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1320mm) | Midspan s<=d/2: s=102mm, d/2=288mm (5a-5a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3269 | 2302 | PASS | 50 | 50 | PASS | 3269 | - | 1320 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 852 | 426 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 852 | 426 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 852 | 213 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 852 | 213 | PASS |

## LONGITUDINAL SECTION OF GRADE BEAM-GB1(14"x26"): ALONG GRID-7 3/7" = 1'-0"

GB1 | 356x660mm (bxh) | 3 span(s) | 8 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 6a-6a | 660 | 577.5 | d10 | 102 | 144 | PASS | 2/2 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1320mm) | Midspan s<=d/2: s=102mm, d/2=289mm (6a-6a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2962 | 2310 | PASS | 50 | 50 | PASS | 2962 | - | 1320 | PASS |
| 2 | 3564 | 2310 | PASS | 50 | 50 | PASS | 1200 | 1193 | 1320 | FAIL |
| 3 | 3668 | 2310 | PASS | 50 | 100 | PASS | 1200 | 1200 | 1320 | FAIL |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 568 | 284 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 568 | 284 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 142 | PASS |

| | | | | |
|---|---|---|---|---|
| 1 | min Mn+ along span >= max joint As/4 | 568 | 142 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 568 | 284 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 568 | 284 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 568 | 142 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 568 | 142 | PASS |
| 3 | Mn,l+ >= Mn,l-/2 | 568 | 284 | PASS |
| 3 | Mn,r+ >= Mn,r-/2 | 568 | 284 | PASS |
| 3 | min Mn- along span >= max joint As/4 | 568 | 142 | PASS |
| 3 | min Mn+ along span >= max joint As/4 | 568 | 142 | PASS |

## LONGITUDINAL SECTION OF GRADE BEAM-GB2(14"x26"): ALONG GRID-A 3/7" = 1'-0"

GB2 | 356x660mm (bxh) | 2 span(s) | 6 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 7a-7a | 660 | 577.5 | d10 | 102 | 144 | PASS | 2/2 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1320mm) | Midspan s<=d/2: s=102mm, d/2=289mm (7a-7a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4587 | 2310 | PASS | 50 | 50 | PASS | 1309 | 1309 | 1320 | FAIL |
| 2 | 3706 | 2310 | PASS | 50 | 50 | PASS | 1209 | 1209 | 1320 | FAIL |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 568 | 284 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 568 | 284 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 142 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 568 | 142 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 568 | 284 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 568 | 284 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 568 | 142 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 568 | 142 | PASS |

## LONGITUDINAL SECTION OF GRADE BEAM-GB2(14"x26"): ALONG GRID-B 3/7" = 1'-0"

GB2 | 356x660mm (bxh) | 2 span(s) | 6 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 8a-8a | 660 | 577.5 | d10 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1320mm) | Midspan s<=d/2: s=102mm, d/2=289mm (8a-8a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4587 | 2310 | PASS | 50 | 50 | PASS | 1309 | 1309 | 1320 | FAIL |
| 2 | 3156 | 2310 | PASS | 50 | 50 | PASS | 3156 | - | 1320 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 852 | 426 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 852 | 426 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 852 | 213 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 852 | 213 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 852 | 426 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 852 | 426 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 852 | 213 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 852 | 213 | PASS |

## LONGITUDINAL SECTION OF GRADE BEAM-GB2(14"x26"): ALONG GRID-D 3/7" = 1'-0"

GB2 | 356x660mm (bxh) | 2 span(s) | 6 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 10a-10a | 660 | 577.5 | d10 | 102 | 128 | PASS | 3/2 | PASS | FAIL | FAIL |
| 10b-10b | 660 | 577.5 | d10 | 152 | 128 | FAIL | 2/2 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1320mm) | Midspan s<=d/2: s=152mm, d/2=289mm (10b-10b) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4612 | 2310 | PASS | 50 | 50 | PASS | 1209 | 1209 | 1320 | FAIL |
| 2 | 2258 | 2310 | FAIL | 50 | 50 | PASS | 2258 | - | 1320 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 568 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 568 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 568 | 192 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 568 | 384 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 568 | 384 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 568 | 192 | PASS |

## LONGITUDINAL SECTION OF GRADE BEAM-GB2(14"x26"): ALONG GRID-E 3/7" = 1'-0"

GB2 | 356x660mm (bxh) | 4 span(s) | 10 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 11a-11a | 660 | 579.5 | d10 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |
| 11b-11b | 660 | 579.5 | d10 | 152 | 128 | FAIL | 3/3 | PASS | FAIL | FAIL |

Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1320mm) | Midspan s<=d/2: s=152mm, d/2=290mm (11b-11b) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4612 | 2318 | PASS | 50 | 50 | PASS | 1209 | 1209 | 1320 | FAIL |
| 2 | 2258 | 2318 | FAIL | 50 | 50 | PASS | 2258 | - | 1320 | PASS |
| 3 | 1829 | 2318 | FAIL | 50 | 50 | PASS | 1829 | - | 1320 | PASS |
| 4 | 2502 | 2318 | PASS | 50 | 50 | PASS | 2502 | - | 1320 | PASS |

Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 768 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 3 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 3 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 3 | min Mn- along span >= max joint As/4 | 768 | 192 | PASS |
| 3 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 4 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 4 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 4 | min Mn- along span >= max joint As/4 | 768 | 192 | PASS |
| 4 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-FB1(12"X24"): ALONG GRID-1 3/7" = 1'-0"

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 550.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |

Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=275mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1977 | 2202 | FAIL | 50 | 50 | PASS | 1977 | - | 1220 | PASS |

Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 852 | 426 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 852 | 426 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 852 | 213 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 852 | 213 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-FB1(12''x24''): ALONG GRID-2 3/7'' = 1'-0''

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 2a-2a | 610 | 552.5 | d10 | 102 | 128 | PASS | 2/3 | PASS | FAIL | FAIL |
| 2b-2b | 610 | 552.5 | d10 | 102 | 128 | PASS | 2/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (2a-2a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1772 | 2210 | FAIL | 50 | 50 | PASS | 1772 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 852 | 284 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 852 | 284 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 213 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 852 | 213 | PASS |

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## LONGITUDINAL SECTION OF FLOOR BEAM-FB1(12''x24''): IN BETWEEN GRID-B~C 3/7'' = 1'-0''

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 3a-3a | 610 | 550.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=275mm (3a-3a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1335 | 2202 | FAIL | 50 | 50 | PASS | 1335 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 852 | 426 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 852 | 426 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 1252 | 213 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 852 | 213 | PASS |

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## LONGITUDINAL SECTION OF FLOOR BEAM-FB1(12''X24''): ALONG GRID-3 3/7'' = 1'-0''

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 3a-3a | 610 | 550.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=275mm (3a-3a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1335 | 2202 | FAIL | 50 | 50 | PASS | 1335 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 852 | 426 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 852 | 426 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 1252 | 213 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 852 | 213 | PASS |

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## LONGITUDINAL SECTION OF FLOOR BEAM-FB1(12''x24''): ALONG GRID-4 3/7'' = 1'-0''

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 5a-5a | 610 | 552.5 | d10 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (5a-5a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2212 | 2210 | PASS | 50 | 50 | PASS | 2212 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 852 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 852 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 768 | 213 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 852 | 213 | PASS |

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## LONGITUDINAL SECTION OF FLOOR BEAM-FB1(12''x24''): ALONG GRID-5 3/7'' = 1'-0''

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 5a-5a | 610 | 552.5 | d10 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (5a-5a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2212 | 2210 | PASS | 50 | 50 | PASS | 2212 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 852 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 852 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 768 | 213 | PASS |

| | | | | |
|---|---|---|---|---|
| 1 | min Mn+ along span >= max joint As/4 | 852 | 213 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-FB1(12''x24''): ALONG GRID-6 3/7'' = 1'-0''

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 6a-6a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (6a-6a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3269 | 2210 | PASS | 50 | 50 | PASS | 3269 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 768 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-2FB1(12''x24''): ALONG GRID-7 3/7'' = 1'-0''

FB1 | 305x610mm (bxh) | 3 span(s) | 8 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 7a-7a | 610 | 554.5 | d10 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |
| 7b-7b | 610 | 554.5 | d10 | 102 | 128 | PASS | 2/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=277mm (7a-7a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2962 | 2218 | PASS | 50 | 50 | PASS | 2962 | - | 1220 | PASS |
| 2 | 3564 | 2218 | PASS | 50 | 50 | PASS | 1200 | 1193 | 1220 | FAIL |
| 3 | 3668 | 2218 | PASS | 50 | 100 | PASS | 1200 | 1200 | 1220 | FAIL |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

| | | | | |
|---|---|---|---|---|
| 3 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 3 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 3 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 3 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-2FB2(12"x24"): ALONG GRID-A 3/7" = 1'-0"

FB2 | 305x610mm (bxh) | 2 span(s) | 6 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 8a-8a | 610 | 550.5 | d12 | 102 | 128 | PASS | 3/2 | PASS | FAIL | FAIL |
| 8b-8b | 610 | 550.5 | d12 | 152 | 128 | FAIL | 3/2 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=152mm, d/2=275mm (8a-8a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4587 | 2202 | PASS | 50 | 50 | PASS | 1209 | 1209 | 1220 | FAIL |
| 2 | 3706 | 2202 | PASS | 50 | 50 | PASS | 1195 | 1195 | 1220 | FAIL |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 568 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 568 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 1336 | 150 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 568 | 300 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 568 | 300 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 1336 | 150 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-2FB2(12"x24"): ALONG GRID-B 3/7" = 1'-0"

FB2 | 305x610mm (bxh) | 2 span(s) | 6 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 9a-9a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |
| 9b-9b | 610 | 552.5 | d12 | 152 | 128 | FAIL | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=152mm, d/2=276mm (9a-9a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4587 | 2210 | PASS | 50 | 50 | PASS | 1209 | 1209 | 1220 | FAIL |
| 2 | 3156 | 2210 | PASS | 50 | 50 | PASS | 3156 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 768 | 300 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 768 | 300 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 600 | 192 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-2FB2(12"x24"): ALONG GRID-C 3/7" = 1'-0"

FB2 | 305x610mm (bxh) | 2 span(s) | 6 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 9a-9a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |
| 9b-9b | 610 | 552.5 | d12 | 152 | 128 | FAIL | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=152mm, d/2=276mm (9a-9a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4587 | 2210 | PASS | 50 | 50 | PASS | 1209 | 1209 | 1220 | FAIL |
| 2 | 3156 | 2210 | PASS | 50 | 50 | PASS | 3156 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 768 | 300 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 768 | 300 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 600 | 192 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-2FB2(12"x24"): ALONG GRID-D 3/7" = 1'-0"

FB2 | 305x610mm (bxh) | 2 span(s) | 6 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 11a-11a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |
| 11b-11b | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (11b-11b) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4612 | 2210 | PASS | 50 | 50 | PASS | 1209 | 1209 | 1220 | FAIL |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 2250 | 2210 | PASS | 50 | 50 | PASS | 2250 | - | 1220 | PASS |

#### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 768 | 300 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 768 | 300 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 600 | 192 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

### LONGITUDINAL SECTION OF FLOOR BEAM-2FB2(12''x24''): IN BETWEEN GRID 4~6 3/7" = 1'-0"

FB2 | 305x610mm (bxh) | 1 span(s) | 3 supports

#### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 12a-12a | 610 | 552.5 | d10 | 102 | 128 | PASS | 2/2 | PASS | FAIL | FAIL |

#### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (12a-12a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1825 | 2210 | FAIL | 50 | 50 | PASS | 1825 | - | 1220 | PASS |

#### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 568 | 284 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 568 | 284 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 142 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 568 | 142 | PASS |

### LONGITUDINAL SECTION OF FLOOR BEAM-2FB2(12''x24''): IN BETWEEN GRID 3~4 3/7" = 1'-0"

FB2 | 305x610mm (bxh) | 2 span(s) | 4 supports

#### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 13a-13a | 610 | 550.5 | d12 | 102 | 128 | PASS | 2/2 | PASS | FAIL | FAIL |

#### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=275mm (13a-13a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1534 | 2202 | FAIL | 50 | 50 | PASS | - | - | 1220 | N/A |
| 2 | 2250 | 2202 | PASS | - | - | - | 2250 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | N/A | N/A | N/A |
| 1 | Mn,r+ >= Mn,r-/2 | N/A | N/A | N/A |
| 1 | min Mn- along span >= max joint As/4 | N/A | N/A | N/A |
| 1 | min Mn+ along span >= max joint As/4 | N/A | N/A | N/A |
| 2 | Mn,l+ >= Mn,l-/2 | 568 | 284 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 568 | 284 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 568 | 142 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 568 | 142 | PASS |

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## LONGITUDINAL SECTION OF FLOOR BEAM-2FB1(12"x24"): ALONG GRID-E 3/7" = 1'-0"

FB1 | 305x610mm (bxh) | 4 span(s) | 10 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 14a-14a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |
| 14b-14b | 610 | 552.5 | d12 | 152 | 128 | FAIL | 2/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=152mm, d/2=276mm (14b-14b) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4612 | 2210 | PASS | 50 | 50 | PASS | 1209 | 1209 | 1220 | FAIL |
| 2 | 2258 | 2210 | PASS | 50 | 50 | PASS | 2258 | - | 1220 | PASS |
| 3 | 1829 | 2210 | FAIL | 50 | 50 | PASS | 1829 | - | 1220 | PASS |
| 4 | 2502 | 2210 | PASS | 50 | 50 | PASS | 2502 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 3 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 3 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 3 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 3 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |
| 4 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 4 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 4 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |

| | | | | |
|---|---|---|---|---|
| 4 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-2FB1(12"X24"): ALONG GRID-1 3/7" = 1'-0"

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1977 | 2210 | FAIL | 50 | 50 | PASS | 1977 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-2FB1(12"x24"): ALONG GRID-2 3/7" = 1'-0"

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 2a-2a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |
| 2b-2b | 610 | 552.5 | d12 | 152 | 128 | FAIL | 2/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (2a-2a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1772 | 2210 | FAIL | 50 | 50 | PASS | 1772 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-2FB1(12"x24"): IN BETWEEN GRID-B~C 3/7" = 1'-0"

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 3a-3a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (3a-3a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1335 | 2210 | FAIL | 50 | 50 | PASS | 1335 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 426 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 426 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 852 | 213 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 213 | PASS |

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## LONGITUDINAL SECTION OF FLOOR BEAM-2FB1(12"X24"): ALONG GRID-3 3/7" = 1'-0"

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 3a-3a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | PASS | PASS |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (3a-3a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1335 | 2210 | FAIL | 50 | 50 | PASS | 1335 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 426 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 426 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 852 | 213 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 213 | PASS |

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## LONGITUDINAL SECTION OF FLOOR BEAM-2FB1(12"x24"): ALONG GRID-4 3/7" = 1'-0"

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 5a-5a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (5a-5a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2212 | 2210 | PASS | 50 | 50 | PASS | 2212 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-2FB1(12''x24''): ALONG GRID-5 3/7'' = 1'-0''

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 5a-5a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (5a-5a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2212 | 2210 | PASS | 50 | 50 | PASS | 2212 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 768 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 768 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 192 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 768 | 192 | PASS |

## LONGITUDINAL SECTION OF FLOOR BEAM-2FB1(12''x24''): ALONG GRID-6 3/7'' = 1'-0''

FB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 6a-6a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (6a-6a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3269 | 2210 | PASS | 50 | 50 | PASS | 3269 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 600 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 600 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 768 | 192 | PASS |

| | | | | |
|---|---|---|---|---|
| 1 | min Mn+ along span >= max joint As/4 | 600 | 192 | PASS |

## LONGITUDINAL SECTION OF GRADE BEAM-GB2(14"x26"): ALONG GRID-C 3/7" = 1'-0"

GB2 | 356x660mm (bxh) | 2 span(s) | 6 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 9a-9a | 660 | 577.5 | d10 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |
| 9b-9b | 660 | 577.5 | d10 | 102 | 128 | PASS | 2/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1320mm) | Midspan s<=d/2: s=102mm, d/2=289mm (9a-9a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1695 | 2310 | FAIL | 51 | 50 | PASS | 1694 | - | 1320 | PASS |
| 2 | 1102 | 2310 | FAIL | 50 | 50 | PASS | 1102 | - | 1320 | FAIL |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 852 | 384 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 852 | 384 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 568 | 213 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 852 | 213 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 852 | 384 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 852 | 384 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 568 | 213 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 852 | 213 | PASS |

## LONGITUDINAL SECTION OF LIFT BOTTOM BEAM-LB1(12"x24"): ALONG GRID-6 3/7" = 1'-0"

LB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1150 | 2210 | FAIL | 50 | 50 | PASS | 1150 | - | 1220 | FAIL |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |

## LONGITUDINAL SECTION OF LIFT BOTTOM BEAM-LB1(12"x24"): ALONG GRID-4 3/7" = 1'-0"

LB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2343 | 2210 | PASS | - | - | - | - | - | 1220 | N/A |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |

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## LONGITUDINAL SECTION OF LIFT BOTTOM BEAM-LB1(12"x24"): ALONG GRID-E & IN BETWEEN GRID 4~6 3/7'...

LB1 | 305x610mm (bxh) | 1 span(s) | 5 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1815 | 2210 | FAIL | 50 | 50 | PASS | 1815 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |

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## LONGITUDINAL SECTION OF STAIR TOP BEAM-SB2(12"x24"): ALONG GRID-E 3/7" = 1'-0"

SB2 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

#### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2502 | 2210 | PASS | 50 | 50 | PASS | 2502 | - | 1220 | PASS |

#### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |

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### LONGITUDINAL SECTION OF STAIR TOP BEAM-SB2(12''x24''): ALONG GRID-C 3/7'' = 1'-0''

SB2 | 305x610mm (bxh) | 1 span(s) | 4 supports

#### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

#### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1105 | 2210 | FAIL | 50 | 50 | PASS | 1105 | - | 1220 | FAIL |

#### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |

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### LONGITUDINAL SECTION OF STAIR TOP BEAM-SB1(12''x24''): ALONG GRID-7 3/7'' = 1'-0''

SB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

#### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

#### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3668 | 2210 | PASS | 50 | 50 | PASS | 1200 | 1200 | 1220 | FAIL |

#### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |

| | | | | |
|---|---|---|---|---|
| 1 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |

## LONGITUDINAL SECTION OF STAIR TOP BEAM-SB1(12''x24''): ALONG GRID-7 3/7'' = 1'-0''

SB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3269 | 2210 | PASS | 50 | 50 | PASS | 3269 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |

## LONGITUDINAL SECTION OF O.H.W.T BOTTOM BEAM-WB1(12''x24''): ALONG GRID-7 3/7'' = 1'-0''

WB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3668 | 2210 | PASS | 50 | 50 | PASS | 1200 | 1200 | 1220 | FAIL |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |

## LONGITUDINAL SECTION OF O.H.W.T BOTTOM BEAM-WB1(12''x24''): ALONG GRID-6 3/7'' = 1'-0''

WB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3269 | 2210 | PASS | 50 | 50 | PASS | 3269 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |

## LONGITUDINAL SECTION OF O.H.W.T BOTTOM BEAM-WB1(12''x24''): ALONG GRID-4 3/7'' = 1'-0''

WB1 | 305x610mm (bxh) | 1 span(s) | 4 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2343 | 2210 | PASS | - | - | - | - | - | 1220 | N/A |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |

## LONGITUDINAL SECTION OF O.H.W.T BOTTOM BEAM-WB2(12''x24''): ALONG GRID-C 3/7'' = 1'-0''

WB2 | 305x610mm (bxh) | 2 span(s) | 6 supports

### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1829 | 2210 | FAIL | 50 | 50 | PASS | 1829 | - | 1220 | PASS |
| 2 | 2502 | 2210 | PASS | 50 | 50 | PASS | 2502 | - | 1220 | PASS |

### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |

| | | | | |
|---|---|---|---|---|
| 1 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |

### LONGITUDINAL SECTION OF O.H.W.T BOTTOM BEAM-WB2(12''x24''): ALONG GRID-E 3/7" = 1'-0"

WB2 | 305x610mm (bxh) | 2 span(s) | 6 supports

#### Cross-section checks (stirrup spacing, bar count, rho)

| Section | h | d | Stirrup | s | s_max | Spacing | Top/Bot bars | Bars>=2 | rho | Overall |
|---|---|---|---|---|---|---|---|---|---|---|
| 1a-1a | 610 | 552.5 | d12 | 102 | 128 | PASS | 3/3 | PASS | FAIL | FAIL |

#### Span checks: ln>=4d, 1st stirrup>=50mm, zone length>=2h (1220mm) | Midspan s<=d/2: s=102mm, d/2=276mm (1a-1a) => PASS

| Span | ln | 4d | ln>=4d | 1st stir L | 1st stir R | >=50mm | Zone L | Zone R | 2h | Zone>=2h |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1829 | 2210 | FAIL | 50 | 50 | PASS | 1829 | - | 1220 | PASS |
| 2 | 2502 | 2210 | PASS | 50 | 50 | PASS | 2502 | - | 1220 | PASS |

#### Flexural reinforcement area ratio checks

| Span | Rule | As prov (mm2) | As req (mm2) | Result |
|---|---|---|---|---|
| 1 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 1 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |
| 1 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 1 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |
| 2 | Mn,l+ >= Mn,l-/2 | 600 | 300 | PASS |
| 2 | Mn,r+ >= Mn,r-/2 | 600 | 300 | PASS |
| 2 | min Mn- along span >= max joint As/4 | 600 | 150 | PASS |
| 2 | min Mn+ along span >= max joint As/4 | 600 | 150 | PASS |

## Parsing notes (157)

- LONGITUDINAL SECTION OF FLOOR BEAM-FB1(1 | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-FB1(1 | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-FB2(1 | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-FB2(1 | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-FB2(1 | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-FB2(1 | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-FB2(1 | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-FB1(1 | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF GRADE BEAM-GB1(1 | Stirrup zone length < 2h (1320mm)
- LONGITUDINAL SECTION OF GRADE BEAM-GB1(1 | Stirrup zone length < 2h (1320mm)
- LONGITUDINAL SECTION OF GRADE BEAM-GB1(1 | Stirrup zone length < 2h (1320mm)
- LONGITUDINAL SECTION OF GRADE BEAM-GB1(1 | Stirrup zone length < 2h (1320mm)
- LONGITUDINAL SECTION OF GRADE BEAM-GB2(1 | Stirrup zone length < 2h (1320mm)
- LONGITUDINAL SECTION OF GRADE BEAM-GB2(1 | Stirrup zone length < 2h (1320mm)
- LONGITUDINAL SECTION OF GRADE BEAM-GB2(1 | Stirrup zone length < 2h (1320mm)

- LONGITUDINAL SECTION OF GRADE BEAM-GB2(1 | Stirrup zone length < 2h (1320mm)
- LONGITUDINAL SECTION OF GRADE BEAM-GB2(1 | Stirrup zone length < 2h (1320mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-2FB1( | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-2FB1( | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-2FB2( | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-2FB2( | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-2FB2( | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-2FB2( | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-2FB2( | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF FLOOR BEAM-2FB1( | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF GRADE BEAM-GB2(1 | Stirrup zone length < 2h (1320mm)
- LONGITUDINAL SECTION OF LIFT BOTTOM BEAM | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF STAIR TOP BEAM-S | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF STAIR TOP BEAM-S | Stirrup zone length < 2h (1220mm)
- LONGITUDINAL SECTION OF O.H.W.T BOTTOM B | Stirrup zone length < 2h (1220mm)
- sec A1-A1 | No rebar circles found
- sec A-A | No rebar circles found
- sec C-C | No rebar circles found
- sec 1a-1a | Top reinforcement ratio is outside permitted limits
- sec 1a-1a | Bottom reinforcement ratio is outside permitted limits
- sec 1b-1b | Top reinforcement ratio is outside permitted limits
- sec 1b-1b | Bottom reinforcement ratio is outside permitted limits
- sec 2a-2a | Top reinforcement ratio is outside permitted limits
- sec 2a-2a | Bottom reinforcement ratio is outside permitted limits
- sec 2b-2b | Stirrup spacing exceeds maximum allowed (support/end section)